

Work through the formula step-

by-step to find the answer to the equation. First simplify the values under the square root sign. Work out the square of 3 (which equals 9), then work out the value of $4 \times 1 \times -2$ (which equals -8).

$$x = \frac{-3 \pm \sqrt{9 - (-8)}}{2}$$

$3 \times 3 = 9$
 $4 \times 1 \times (-2) = -8$
two minus signs cancel out, so $9 - (-8) = 9 + 8$

Work out the numbers under the square root sign: $9 - (-8)$ equals $9 + 8$, which equals 17. Then, use a calculator to find the square root of 17.

$$x = \frac{-3 \pm \sqrt{17}}{2}$$

$9 + 8 = 17$

Once the sum is simplified, it must be split to find the two answers—one when the second value is subtracted from the first, and the other where they are added.

$$x = \frac{-3 + 4.12}{2} \quad x = \frac{-3 - 4.12}{2}$$

4.12 is the square root of 17 rounded to 2 decimal places.

Add the two values on the top part of the fraction; here the values are -3 and 4.12 .

$$x = \frac{-3 + 4.12}{2}$$

Subtract the second value on the top part of the fraction from the first value; here the values are -3 and -4.12 .

$$x = \frac{-3 - 4.12}{2}$$

Divide the top part of the fraction by the bottom part to find an answer.

$$x = \frac{1.12}{2}$$

$-3 + 4.12 = 1.12$

Divide the top part of the fraction by the bottom part to find an answer.

$$x = \frac{-7.12}{2}$$

$-3 - 4.12 = -7.12$

Give both answers, because quadratic equations always have two solutions.

$$x = 0.56$$

quadratic equations always have two solutions

$$x = -3.56$$